

Programming with C - Assignment 2

Total Marks: 100

Due Date: 10/04/2026

Question 1: Student Attendance Tracker (25 Points)

Write a C program that:

- Reads the **total number of classes held** (integer) and the **number of classes attended** by a student (integer)
- Uses an **if-else if-else ladder** to classify attendance:
 - **Excellent:** attendance percentage $\geq 90\%$
 - **Satisfactory:** attendance percentage $\geq 75\%$ and $< 90\%$
 - **Low:** attendance percentage $\geq 60\%$ and $< 75\%$
 - **Critical:** attendance percentage $< 60\%$
- Uses a **switch statement** to print a corresponding advisory message based on classification:
 - Excellent $\rightarrow$ *"Keep it up!"*
 - Satisfactory $\rightarrow$ *"Maintain your attendance."*
 - Low $\rightarrow$ *"Attendance improvement required."*
 - Critical $\rightarrow$ *"You may be barred from exams."*
- Uses a **while loop** to repeatedly accept new attendance inputs until the user enters -1 to quit
- Validates input using branching — if classes attended $>$ total classes, display an error and prompt again
- Displays the attendance percentage formatted to **2 decimal places**

Question 2: Multiplication Table Generator (25 Points)

Write a C program that:

- Reads an integer n from the user (the table limit, e.g., up to the 10th multiple)
- Uses a **for loop** to print the multiplication table of every number from **1 to n**
- Uses a **do-while loop** to re-prompt the user if n is not between 1 and 20 (input validation)
- Uses **nested for loops** to print a **full $n \times n$ multiplication grid**, with values aligned in columns
- Uses an **if-else** inside the loop to mark values that are **perfect squares** with an asterisk (*)

- Uses a **break statement** to stop printing any row once a product exceeds 100
- Uses a **continue statement** to skip printing multiples that are divisible by both 3 and 5

Question 3: Array-Based Sales Analysis (25 Points)

Write a C program that:

- Declares an integer array to store **daily sales figures for 7 days** (one week)
- Reads the 7 sales values from the user using a **for loop**
- Uses loops and branching to compute and display:
 - **Total sales** for the week
 - **Average daily sales**
 - **Maximum and minimum** sales day (display the day number and value)
 - **Number of days** sales were above average
- Uses a **for loop with an if-else** to classify each day:
 - *"High"* if sales > average
 - *"Average"* if sales == average
 - *"Low"* if sales < average
- Uses **bubble sort** (with nested loops and if statements) to sort and display the sales figures in **ascending order**
- Uses a **second array** to store only the sales values that are **above average**, then displays this sub-array

Question 4: String Operations Menu (25 Points)

Write a C program that:

- Reads a string (sentence) from the user with a maximum length of 100 characters
- Presents a **menu using a switch statement** inside a do-while loop with the following options:
 1. Count the **total number of characters** (excluding the null terminator)
 2. Count the number of **vowels and consonants** separately
 3. Count the number of **words** (assume words are separated by single spaces)
 4. **Reverse the string** and display it
 5. Check if the string is a **palindrome** (ignoring spaces and case)
 6. **Convert** the string to uppercase/lowercase (user's choice via a nested if-else)
 7. Exit
- Implements all operations **without using standard string library functions** (strlen, strcpy, etc.) — use loops and character array indexing only
- Uses a **for loop** with branching to process each character

- Validates that the input string is **not empty**; if it is, prompt the user to re-enter using a do-while loop